به نام خدا

Building a Chess Analysis Model

BlunderGuard

From engine evaluations

to learner-facing commentary



Seyed Ali Hosseini Nasab

Supervisor: Dr. Mehrnoush Shamsfard

A score tells us how much—not why



Engine: evaluation + best line

Learner: threat + consequence

Goal: explain the move

The challenge is translating chess evidence into a useful explanation.

Can an LLM explain engine evidence correct?

01

Build

A local analysis pipeline

with engine + language model

02

Adapt

A small model trained

on chess-domain text

03

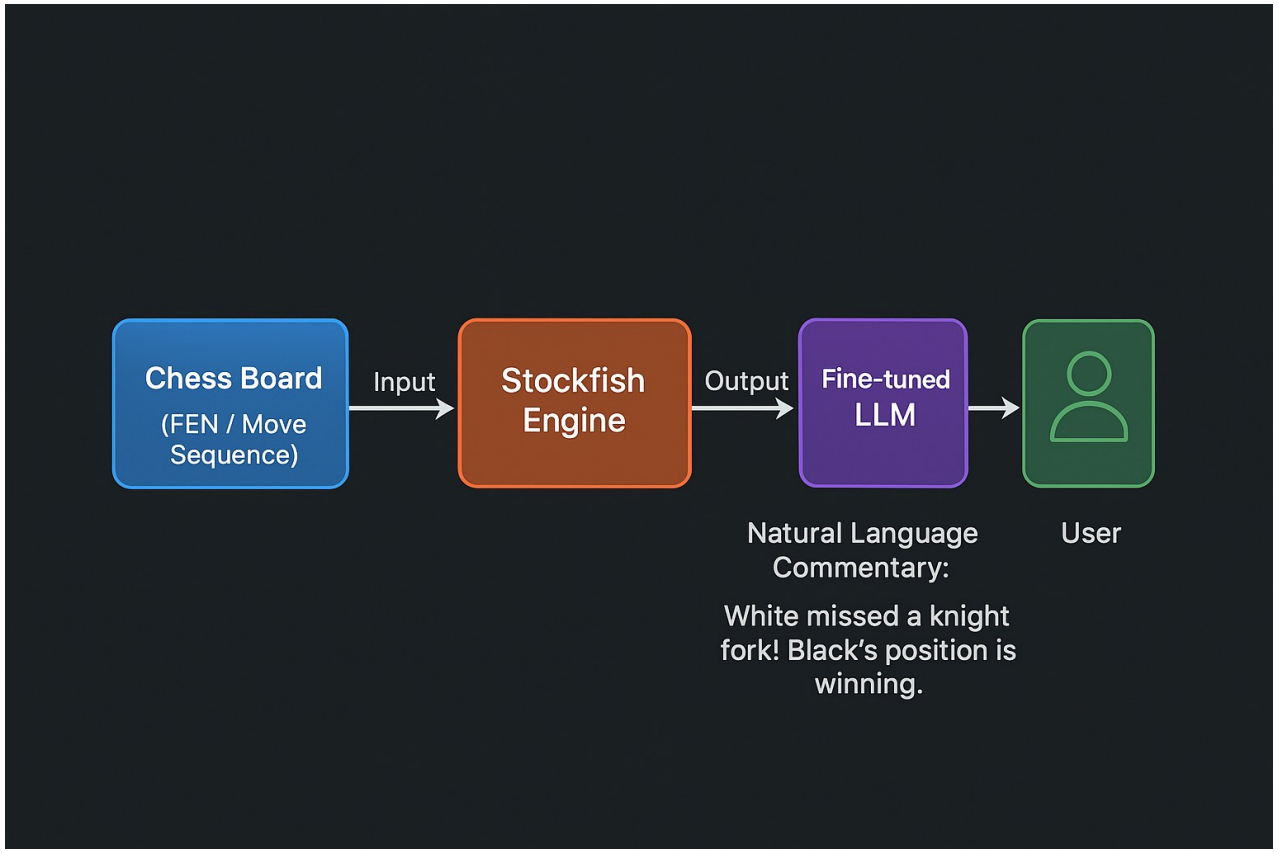
Evaluate

11 models on the same

60 benchmark positions

Success requires chess correctness not just fluent prose.

Engine analysis becomes structured context



Input: position + played move

Stockfish: depth 17, up to 3 lines

Before/after feature differences

LLM: short learner-facing text

Engine evidence guides generation

A small local model: what was actually trained

Base model: Qwen3-1.7B

LoRA adaptation on unpaired chess-book text

Local inference: Q8_0 GGUF with llama.cpp

Domain vocabulary ≠ supervised position-to-explanation learning

One shared benchmark, 660 judgments

60
positions

11
candidate models

660
scored responses

20 positions per rating group: beginner / intermediate / advanced

Candidates aligned by group and row ID

One fixed judge: gemma4-12b; same reference for each position

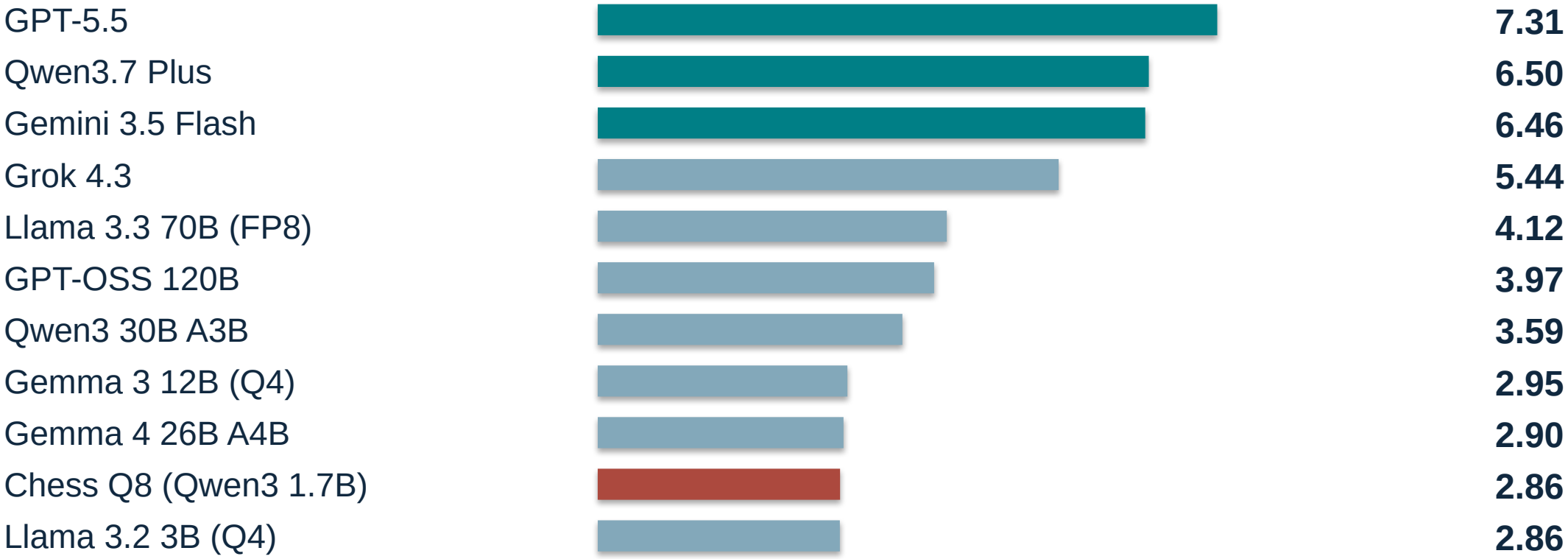
Rating groups describe source games; they are not measured difficulty levels.

The score rewards correctness and faithfulness



The judge compares candidate and reference text; it does not inspect the board.

The results



Mean weighted score / 10 • 60 responses per model

Best observed: GPT-5.5 7.31/10 | Chess Q8 (1.7B): 2.86/10

A 1.7B model scores close to several larger models

Qwen3-1.7B

2.86 / 10

1.7 billion parameters

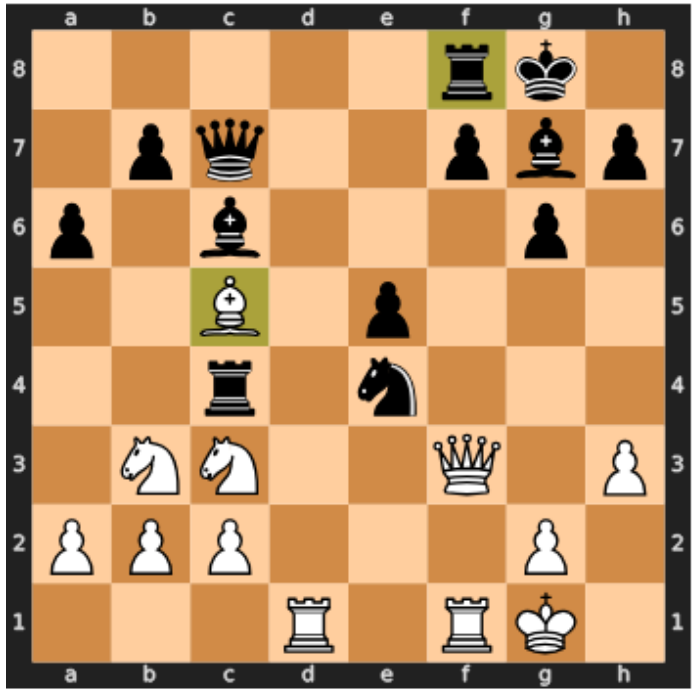
Local GGUF inference

Chess-domain adaptation

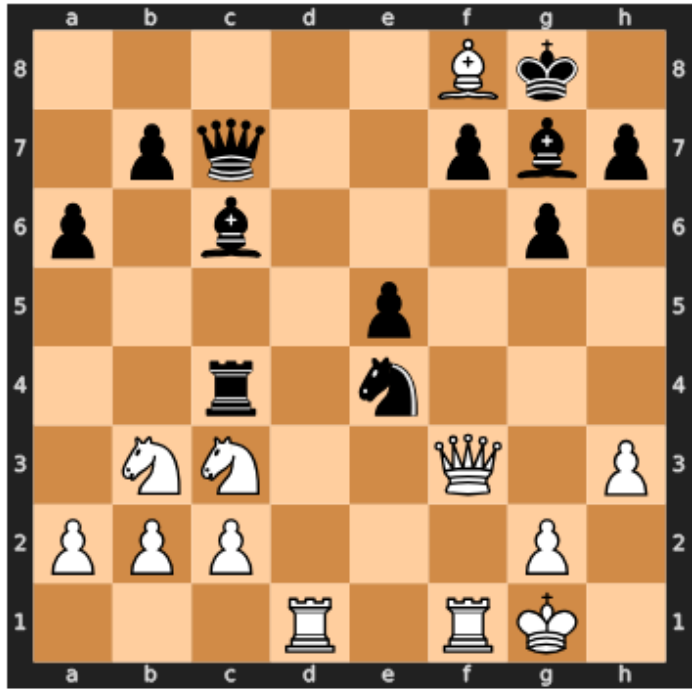


A real failure: the explanation names the wrong move

Before position



After position



This was a blunder move that seriously damaged the position for white.

White's move Nxe4 weakens Black's position by attacking the enemy's king and weakening the knight on c3. Black's move Nxc3 targets the knight, which is a vital piece for Black's defense, and prepares to exploit the weakened position. Both moves control the board and keep the threat alive.

Played move: Bxf8

Model output (excerpt)

“White's move Nxe4 weakens Black's position…”

Correctness 0/10

Faithfulness 0/10

Fluency 10/10

advanced / g1_m38 · overall 2.00/10

From this prototype to a more reliable chess coach

01

Ground the training

Pair board positions with

verified explanations

02

Verify the claims

Check moves and threats

against chess tools

03

Test with people

Expand the benchmark

and add expert review

A small model, a practical starting point

1.7B

parameters · Qwen3

2.86 / 10

mean benchmark score

What this project contributes

An engine-guided commentary pipeline

Scores close to selected larger models

A clear path to improve factual accuracy

Thank you

Questions & discussion